

Technology Plus

Unit 3: Hardware

2.3.1 – Storage Devices

Lesson Overview:

In this lesson, students will identify, describe, and distinguish between volatile and non-volatile storage, and various local storage devices.

Students will:

- Identify and describe volatile and non-volatile storage, and various local storage devices.
- Distinguish between volatile and non-volatile storage, and various local storage devices.
- Evaluate scenarios and determine what local storage device a user needs.

Guiding Question: How can a user identify and distinguish between volatile and non-volatile storage, and various local storage devices?

Suggested Grade Levels: 8-12

Technology Needed: No

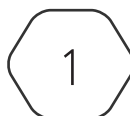
Approximate Time Required: 50-80 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.3 Compare and compare storage types.
 - o Volatile vs. non-volatile
 - o Local storage
 - RAM
 - Read-only memory (ROM)
 - Storage drive
 - Magnetic disks/hard disk drive (HDD)
 - Solid-state drive (SSD)
 - Non-volatile memory express (NVMe)
 - Optical
 - External flash drives
 - o Local network storage
 - Network-attached storage (NAS)
 - File server
 - Cloud storage service

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Lesson Background

Background Information:

In this lesson, students will compare local storage devices and distinguish between volatile and non-volatile storage. Students will review concepts like memory, storage, RAM, ROM, local storage, and storage drives. Students may encounter new concepts such as SATA, volatile, non-volatile, NVMe, and more.

By the end of the lesson students should be able to describe data transfer, storage types, local storage methods, and identify the best storage device for a user based on their needs and preferences.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.3.1 – Storage Devices • Activity Slides 2.3.1 – Storage Devices Scenarios
Materials per Student	• Student Notes 2.3.1 – Storage Devices • Student Handout 2.3.1 – Storage Devices • Assessment 2.3.1 – Storage Devices (optional)

Lesson Background

Cyber Terms:

Term	Definition
serial advanced technology attachment (SATA)	used to connect storage devices to a computer's motherboard
volatile storage	temporary storage that loses data when a system is powered off or restarted
non-volatile storage	permanent storage that retains data without the system needing continuous power
local storage	stores data on physical devices that are directly connected to a network or computer
random access memory (RAM)	a system's short-term memory that provides the temporary storage a CPU needs to perform tasks quickly
read-only memory (ROM)	permanent storage that uses embedded systems to hold firmware and system-level instructions
hard disk drive (HDD)	non-volatile storage that uses spinning disks and magnetic heads to read and write data
solid state drive (SSD)	non-volatile storage that uses flash memory to store data
non-volatile memory express (NVMe)	a protocol used with SSDs to provide faster data transfer speeds through a direct connection to the CPU via PCIe bus
optical drives	non-volatile storage that uses laser technology to read and write data on disks

2.3.1 – Storage Devices

Guiding Question: How can a user identify and distinguish between volatile and non-volatile storage, and various local storage devices?

Lesson Launch (5-10 minutes)

Memory and Storage

- Imagine your friend sending you a gif via text message.
- While you watch the gif, the CPU is storing it in your device's memory (RAM) temporarily.
- You save the gif to your device's camera roll.
- The gif is now permanently saved to your device's storage, and you can watch it directly from your files.
- Memory (RAM) is where our computer temporarily stores programs, data, and the operating system that is being used in real-time.
- Storage is where our computer persistently stores programs, data, and software that is used over long periods of time.

What's Driving Your Data: Storage Devices (20-30 minutes)

- Distribute Student Handout 2.3.1 – Storage Drives and encourage students to fill in the answers during the lesson.

SATA and Data Transfer

- Introduce **SATA**, its definition, and how it impacts data transfer in computers.
 - SATA does not directly connect with the CPU/RAM – it only allows the storage device to communicate with CPU/RAM by connecting it to the motherboard.
 - Hot-swapping is the ability to swap drives while a computer is still powered on. Hot swapping is useful for systems that need to avoid downtime.
 - Motherboards have multiple SATA ports to allow the connection of multiple storage devices and the ability to increase devices as needed.

Volatile vs. Non-Volatile Storage

- Introduce **volatile** and **non-volatile storage**, their definitions, and their uses.
 - Highlight the differences between the two and why it is important that students know if a storage device is volatile or non-volatile.
 - Volatile storage should be used to increase or sustain processing speeds and when the data will not be needed later.
 - Non-volatile storage should be used for data that is important and will need to be

accessed and utilized later.

Local Storage Methods

- Discuss **local storage**, its definition, and the advantages and disadvantages of using local storage.
- Discuss **RAM**, its storage type, definition, and its advantages and disadvantages.
- Discuss **ROM**, its storage type, definition, and its advantages and disadvantages.
 - Some ROM can be rewritten when firmware updates are released.
- Discuss storage drives, their definition, common devices, and their advantages and disadvantages.
 - Storage drives are considered local storage methods; they refer to the actual device that is being used to store data and not the location.
 - Storage drives offer various options from low capacity to high capacity, and slow access speeds to fast access speeds.
- Discuss **HDDs**, their storage type, definition, form factors, and its advantages and disadvantages.
 - Although typically found in computers, some HDDs can be purchased as portable devices and connect to a computer via USB cable.
 - The size of the disks for a hard drive are measured by diameter.
 - Revolutions per minute (RPM) represent the speed at which the magnetic disks spin.
 - The moving parts inside HDDs make them less durable because the parts can be easily broken with harsh, frequent movements.
- Discuss **SSDs**, their storage type, definition, and its advantages and disadvantages.
 - M.2 is a hardware specification (size and shape) for SSDs. Many M.2 drives use NVMe, which is faster than SATA.
- Discuss **NVMe**, its definition, and its advantages and disadvantages.
 - PCIe is faster than the SATA interface - which is mainly used for traditional hard drives or optical drives.
- Discuss **optical drives**, their storage type, definition, and its advantages and disadvantages.
 - Newer technologies can read older storage types, but older storage cannot read newer storage types.
 - i.e. Blu-ray drives can read CDs, but CD drives cannot read Blu-ray disks.
 - CDs can be read-only memory (CD-ROM) or recordable (CD-R/CD-RW) where users can write data or rewrite data on the disk.
 - Blu-rays can store 25GB, DVDs can hold between 4.7GB and 9.4GB, CDs can hold 700MB.
- Discuss **external flash drives**, their storage type, definition, and its advantages and disadvantages.
 - Flash drives have varying types and connection interfaces, such as USB 2.0, USB 3.2, USB-C, and more.

Activity: Storage Devices Scenarios (20-30 minutes)

- Display Activity Slides 2.3.1 – Storage Devices Scenarios, have students write their answers for each scenario at the bottom of the student handout.
- Read each scenario aloud or let students read independently.
- Provide students with time to write their answers.
- Students will determine which type of storage device would best fit a user's needs and why. When choosing a storage device consider performance, permanence (volatile vs. non-volatile), portability, and cost.

Scenarios and Answers

1. Kevin is building a high-end gaming PC and wants the fastest possible load times for his games and operating system.

Kevin needs a NVMe SSD to connect to the PCIe for high data transfer and reduced latency. This will be faster than SATA SSDs.

2. Amira wants her laptop to start up fast and open applications quickly. She doesn't need extreme performance but wants better speed than a hard drive.

Amira needs a SATA SSD because they are faster than HDDs and are durable and reliable for daily use.

3. Emily wants to store her personal photos, videos, and documents on the internal drive of her personal laptop.

Emily needs a local storage device that is attached to her computer, this device could be an internal HDD or SSD able to be accessed without the internet.

4. Ryan is running a graphic design program, editing a large poster, and switching between multiple apps. He notices that when more programs are open, the computer slows down.

Ryan needs to use RAM to temporarily hold data for applications currently in use, it will clear when the computer is off. RAM will free up space for Ryan's CPU to continue processing data at an efficient speed.

5. A history teacher wants to store digital copies of old lectures and class materials on physical media for long-term archiving.

The history teacher needs optical drives for archiving and distributing large amounts of data for a low cost.

6. A local library wants to install low-cost computers for public use that will store lots of eBooks and documents but don't need super-fast performance.

The local library needs HDDs because they offer a large storage capacity at lower costs and have slower performance than other storage types.

7. When Maria turns on her computer, it loads system instructions that never change and allow the operating system to begin booting.

Maria needs ROM to permanently store firmware like BIOS/UEFI which initiates the boot process for a computer.

8. Leo is presenting at a school competition and needs a quick, portable way to bring his slideshow from home to the event.

Leo needs an external flash drive because they are portable and can work with various devices and connect via USB ports.

Putting It All Together (5-10 minutes)

Discussion Questions

- Review what students learned today by discussing the questions below as a class.
- How does SATA impact data transfer?

SATA connects storage devices to the motherboard and allows data to move between the storage device, CPU, and RAM. SATA affects how quickly files load, computers boot, and data is copied.

- What is the difference between volatile and non-volatile storage?

Volatile storage is temporary and loses data when power is off, but it's very fast for short-term use. Non-volatile storage retains data without power and is used for long-term storage.

- What are the differences between RAM and ROM?

RAM is volatile and temporarily stores data and instructions the CPU is using. ROM is non-volatile and permanently stores firmware and system-level instructions.

- What should users consider when choosing a local storage method?

Users should consider performance needs, whether they need volatile or non-volatile storage, cost, capacity, and portability.

- How does NVMe impact SSD storage drives?

NVMe is a protocol that lets SSDs communicate directly with the CPU over PCIe. NVMe SSDs achieve much faster speeds and lower latency than SATA SSDs.